MIDNIGHT VERB

16 BIT DIGITAL EFFECTS PROCESSOR

User Manual

Overview

Midnight Verb models three Alesis rack units from the mid-1980s: the MidiVerb (1985), MidiFex (1986), and MidiVerb II (1986). All 228 factory effect programs are included - reverbs, gates, reverse, delays, chorus, flange and stereo effects - running the original decompiled algorithms. Added on top: pre-delay, tilt EQ, stereo width, input-driven ducking, variable program size, a switchable device clock, and a vintage analog signal chain.

Signal Flow

Modern mode:

Input → Tone EQ → Pre-delay → Anti-alias filter → Effect Algorithm → Width → Ducking → Mix → Output

Vintage mode:

Input → Tone EQ → Pre-delay → 13kHz Resonant LPF → 31Hz HPF → Effect Algorithm → 11kHz Reconstruction LPF → Width → Ducking → Mix → Output

Main Controls

Parameter	Description
Effect	Master bypass switch (Off/On). When Off, the plugin passes dry signal only.
Character	Modern: Clean digital effect core only. Vintage: Adds authentic analog input/output filters based on the MidiVerb II schematic analysis - 13kHz resonant Sallen-Key lowpass (Q=2), 31Hz highpass, and 11kHz reconstruction filter. Subtle but authentic coloration.
Pre-delay	Delay time before the effect begins (0–250ms). Adds separation between dry signal and effect tail. Classic studio technique for clarity on reverbs.
Tone	Adjusts the tonal character of the effect (±10dB tilt EQ at 340Hz–2.7kHz). 50% is neutral. Lower values are darker and warmer, higher values are brighter. Active in both Modern and Vintage modes.
Mix	Wet/dry balance (0–100%). At 100% only the effect signal is output. At 0% the signal is fully dry. Default is 100% for use on send/return buses.
Size	Scales the program's delay times (25–175%). 100% is the hardware's own timing, and sits at the centre of the knob. On the flange and chorus programs it scales the LFO depth instead - the same thing there, since the LFO output is the delay length.
Clock	Runs the emulated device faster or slower (0.5x–2x). Down is longer and darker, up is shorter and brighter. 1x is the hardware rate.
Width	Stereo width of the effect signal (0–200%). 100% leaves it unaltered; 0% collapses it to mono.
LFO Rate	Speed of the internal LFO (10–400%), where 100% is the hardware rate. Active only on MidiVerb II programs 50–69; the control is hidden on every other program.

Ducking Controls

Parameter	Description
Duck Time	Release time for the effect to return after ducking (10–500ms). Shorter times = effect comes back quickly. Longer times = smoother, more gradual swell.
Duck Sens	Sensitivity of the ducking detector (0–100%). Higher values make ducking trigger on quieter signals. Increase when using Midiverb on a send/return bus where input levels are lower.
Duck Amount	Intensity of ducking effect (0–100%). At 0%, ducking is disabled. Higher values duck the effect more aggressively when input signal is present.

Effect Models

MidiVerb (1985) - 64 Programs

The original Alesis MidiVerb was one of the first affordable 16-bit digital reverb processors. All 64 programs are reverb-focused, organized by decay time from 0.2 seconds up to 20 seconds. Includes small, medium, and large room sizes with bright, warm, and dark tonal variations. Programs 51–59 provide gated reverb effects (100–600ms), and programs 60–63 offer reverse reverb (300–600ms). Program 64 is Defeat, which passes the input through unaltered.

MidiFex (1986) - 64 Programs

The Alesis MidiFex was a dedicated delay and effects processor. Its 64 programs cover single echoes (short to long with HPF/BPF/LPF filtering), 2-tap and 3-tap delays with panning and ambience, regenerating delays, slap echoes, reverb hybrids, multitap effects, thickeners, and stereo generation effects. Programs 37–42 (REGEN) are the repeating delays; the ECHO programs give a single repeat. Program 64 is Defeat, which passes the input through unaltered.

MidiVerb II (1986) - 100 Programs

The MidiVerb II expanded on the original with 100 programs across seven categories:

Reverb (Programs 0–29): 30 reverb programs from small bright rooms (0.1s) to extra-large warm halls (15s). Program 0 is Defeat, which passes the input through unaltered.

Gate (Programs 30–39): 10 gated reverb programs with slow and fast gate characteristics (75–450ms).

Reverse (Programs 40–49): 10 reverse reverb and bloom effects (150ms–8s), including regenerating reverse programs.

Flange (Programs 50–59): 10 flange programs including triggered flanges and stereo panning flanges. These use the internal LFO.

Chorus (Programs 60–69): 10 chorus programs from light to deep, plus fast chorus. These use the internal LFO.

Delay (Programs 70–89): 20 delay programs with fixed delay times from 35ms to 460ms.

EFX (Programs 90–99): 10 special effects including multi-tap, stereo generation, thickener, and regenerating delays (2–4s).

Presets

Midnight Verb includes 228 factory presets organized in folders matching the three hardware models. MidiVerb II presets are further organized into subcategories (Reverb, Gate, Reverse, Flange, Chorus, Delay, EFX). Each preset sets the correct model and program number, with all other parameters at default values so you can use them as starting points for your own tweaks. A second bank, Insert 30% Mix, holds the same 228 presets with Mix at 30% for use directly on a track rather than on a send.

Model and Program are set by the presets and are not on the front panel. Both remain available to host automation if you want to sweep programs — Model selects MV-1, MFX or MV-2, and Program runs 0–63 on MV-1 and MFX, 0–99 on MV-2.

Quick Start

Lush hall reverb: Model: MidiVerb II, Program 21 (Large Warm 2.2s), Mix 40–60%, Pre-delay 30–50ms

Small room ambience: Model: MidiVerb, Program 01 (Small Bright 0.2s), Mix 25–40%

Classic 80s gated drums: Model: MidiVerb II, Program 30–39 (Gate), Mix 60–80%

Echo/delay effects: Model: MidiFex, browse programs 01–21 for echo variations

Chorus/flange: Model: MidiVerb II, Program 50–69 (Flange/Chorus), Mix 30–50%

Stereo widening: Model: MidiFex, Program 60–63 (StereoGen), Mix 50–70%

Vintage character: Set Character to Vintage for authentic mid-80s analog coloration

Clean mix with ducking: Any effect, Duck Amount 50–70%, Duck Time 100–200ms

100% wet for send/return: Mix 100% (default). Adjust to taste.

Noise Gate

Midnight Verb includes an automatic noise gate that activates after 16 seconds of silence, fading the wet signal to zero over 6 seconds. This prevents the emulated 16-bit noise floor from being audible during quiet passages. The gate resets immediately when new input signal is detected.

Technical Notes

Midnight Verb runs decompiled effect algorithms from the original Alesis hardware - the same DSP code the MidiVerb, MidiFex and MidiVerb II executed. Each model runs at its own clock, derived the way the hardware derived it: 23.4375 kHz on the MidiVerb and MidiFex (6 MHz ÷ 256 instruction slots per sample) and 31.25 kHz on the MidiVerb II (16 MHz ÷ 2 ÷ 256). Converter depth follows the hardware too - 13-bit on the MidiVerb and MidiFex, 16-bit on the MidiVerb II. Arithmetic is 16-bit integer throughout, over a shared 16K DRAM buffer.

Vintage mode is based on analysis of the MidiVerb II service manual schematics by John Googler Brombaugh, featuring:

• 13kHz resonant Sallen-Key lowpass filter (Q=2) creating an 18dB peak around 8.5kHz

• 31Hz highpass filter for DC blocking

• 11kHz reconstruction lowpass filter on the output

MidiVerb II programs 50–69 (flange and chorus) use an internal LFO with triangle and sine waveforms, updated every 8 samples. Size scales its depth and LFO Rate its speed; at 100% both match the original per-program settings. LFO Rate is hidden on the programs that have no LFO.

Support

For technical support, updates, and additional information:

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